

Individual Assessment Coversheet

To be attached to the front of the assessment.

Campus: Eduvos Mowbray

Faculty: Information Technology

Module Code: ITECA3-12

Group: 1

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Indicate	Yes	No
Plagiarism report attached		

Declaration:

I declare that this assessment is my own original work except for source material explicitly acknowledged. I also declare that this assessment or any other of my original work related to it has not been previously, or is not being simultaneously, submitted for this or any other course. I am aware of the AI policy and acknowledge that I have not used any AI technology to generate or manipulate data, other than as permitted by the assessment instructions. I also declare that I am aware of the Institution's policy and regulations on honesty in academic work as set out in the Conditions of Enrolment, and of the disciplinary guidelines applicable to breaches of such policy and regulations.

Signature nprothschild	Date 5 June 2026
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Deliverable 2 – Rising Tide E-Commerce Platform

Website Link: <https://rising-tide.fredev.app/>

Github Repository: <https://github.com/Noah-Rothschild/rising-tide>

Test buyer: buyer@rising-tide.com **Password:** password

Test seller: clothingco@rising-tide.com **Password:** password

Admin: admin@rising-tide.com **Password:** admin

Introduction

This project presents Rising Tide, a secure consumer-to-consumer (C2C) marketplace designed to be an accessible digital trading platform that focuses on empowering South Africa's informal economy. The core philosophy behind the platform is reflected in the analogy "a rising tide lifts all boats," which highlights the potential for broader economic benefit through improved access to safe digital trade. This solution addresses common issues that exist within other platforms such as Gumtree and Facebook Marketplace where transactions are commonly unverified and rely heavily on unsafe cash payments. These practices expose users to scams and unsafe trading practices.

Rising Tide aims to provide a trustworthy and structured environment for buying and selling goods through user verification, formal delivery and by promoting digital participation of the informal economy. The system is designed to be accessible and responsive to accommodate users with varying digital capabilities. The platform was developed utilising core web technologies such as HTML, CSS, PHP, JavaScript and MySQL. These technologies help to provide a balance between website performance, scalability and interactivity.

Prototyping:

The following figures display the initial user interface of the Rising Tide marketplace including user flow from the home page to marketplace and through until checkout. Responsive design is displayed by utilising developer tools to simulate desktop, mobile and tablet interfaces.

Home Page:

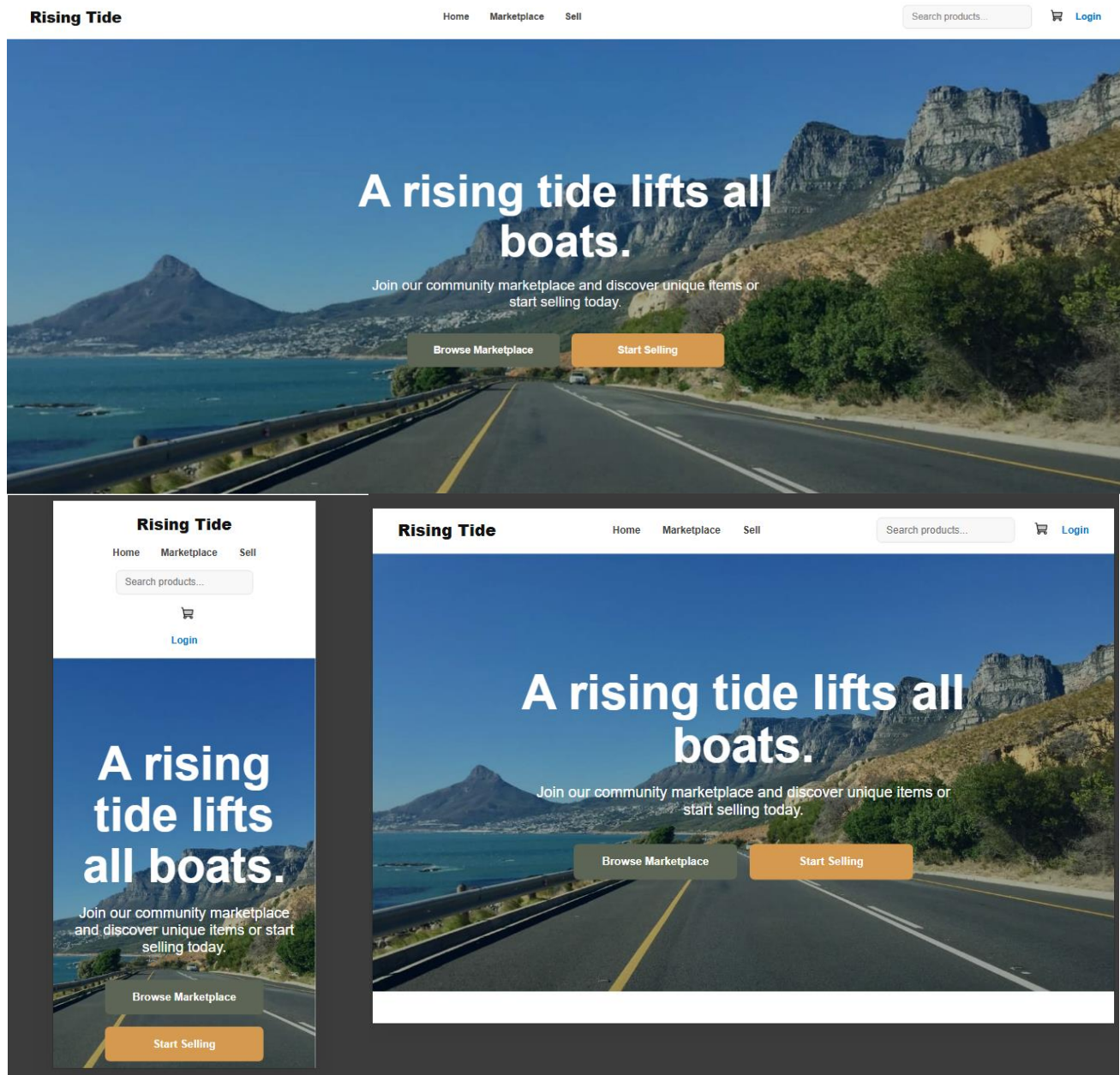


Figure 1: Home Screen including buttons to register as buyer or seller.

Log In Page:

The image displays three versions of the 'Rising Tide' login page. The top view is a large desktop layout. The bottom-left view is a mobile layout with a dark background and a partial image of the boats. The bottom-right view is another mobile layout with a light background and the full image of the boats. All versions share the same core elements: a header with the brand name, navigation links, a search bar, and a login link; a 'WELCOME BACK' heading with a subtext 'Please enter your details.'; input fields for email and password; a 'Remember me' checkbox and a 'Forgot password' link; 'Sign in' and 'Sign in with Google' buttons; and a footer link for new users.

Rising Tide Home Marketplace Sell Search products... Login

WELCOME BACK

Please enter your details.

Email
Enter your email

Password

☐ Remember me [Forgot password](#)

[Sign in](#)

[Sign in with Google](#)

Don't have an account? [Sign up for free!](#)

Figure 2: Login authentication page with links to account registration.

Marketplace Page:

The main landing for the marketplace that provides tools for users to filter down products based on category, price or even through direct search.

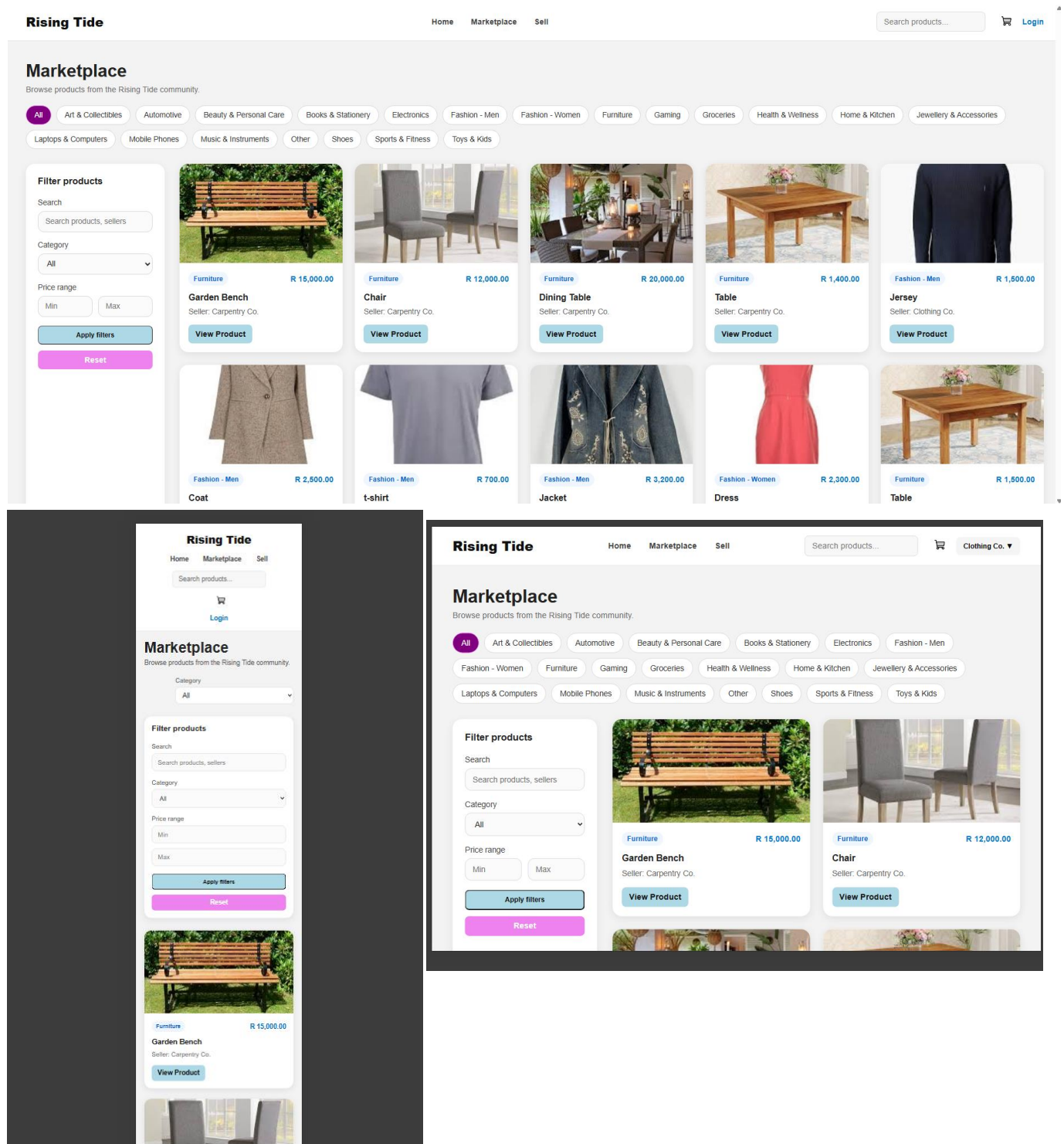


Figure 3: Marketplace page displaying unfiltered product listings.

Product View Page:

A priority was placed on providing users with all the most important information above the fold of the product view page. This includes the product name, price, image, ratings as well as the call-to-action button for 'add to cart'. Longer texts such as product description aren't as important allowing them to be placed lower down on the screen.

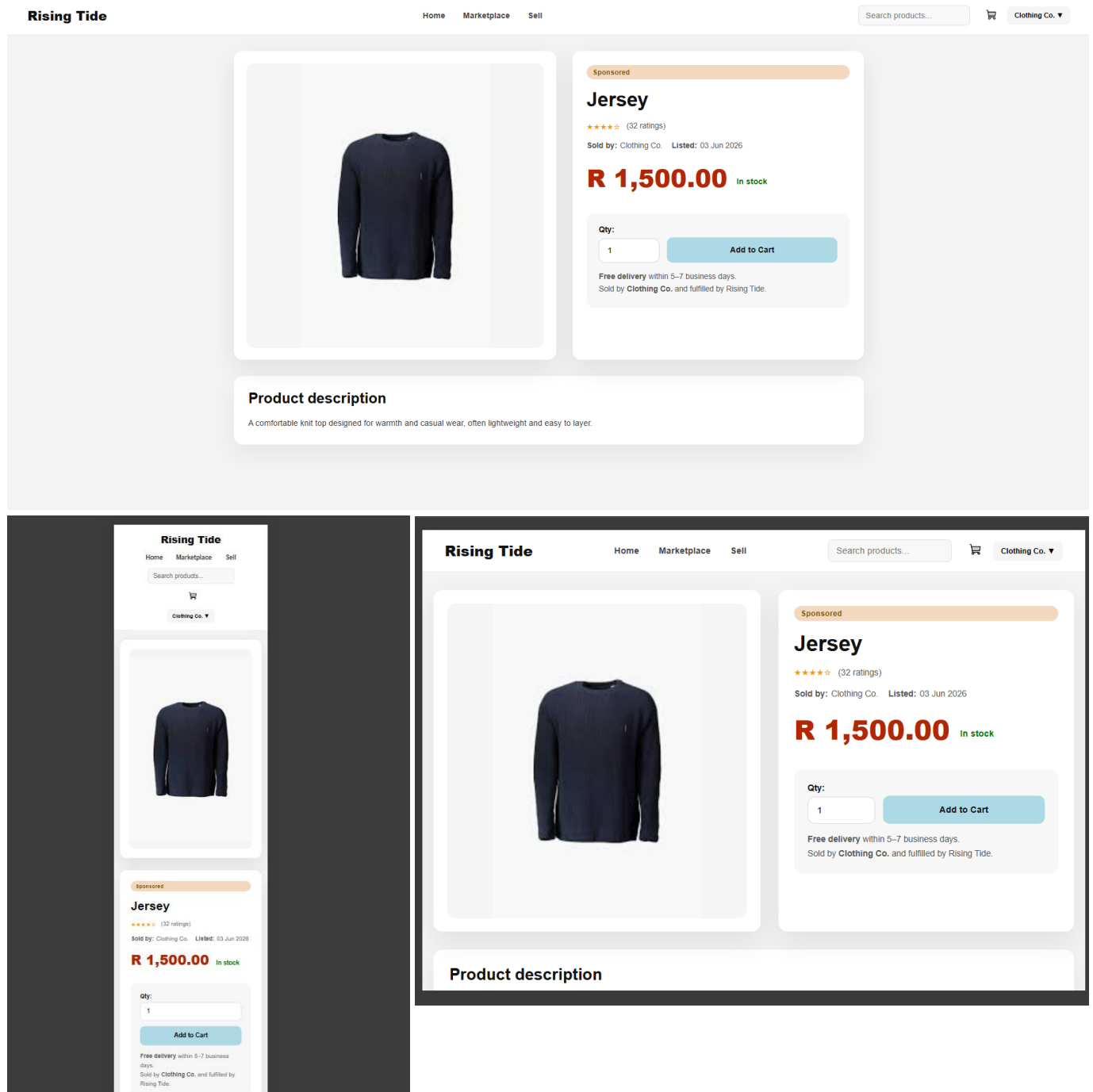


Figure 4: Product View page displaying single jersey product.

Shopping Cart Page:

The shopping cart stores a persistent database of items selected by users up until checkout completes.

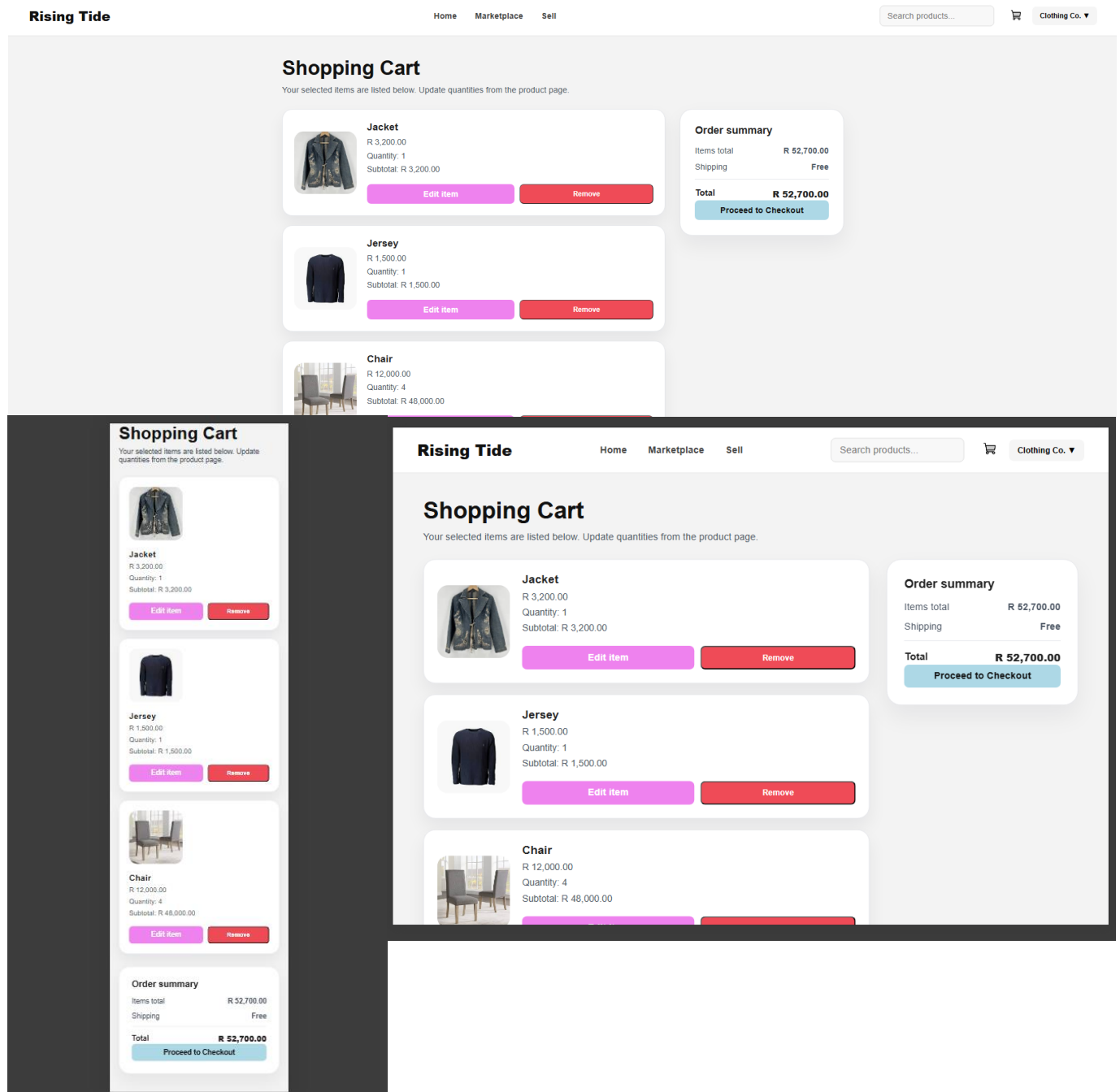


Figure 5: Shopping cart display summary for total calculations before checkout.

Administrator Panel:

The admin dashboard is protected through role-based access permissions to users with the administrator role. Through this panel, administrators can view a list of all users, products and processed orders. Additionally, user roles can be edited, accounts can be deleted alongside problematic product listings.

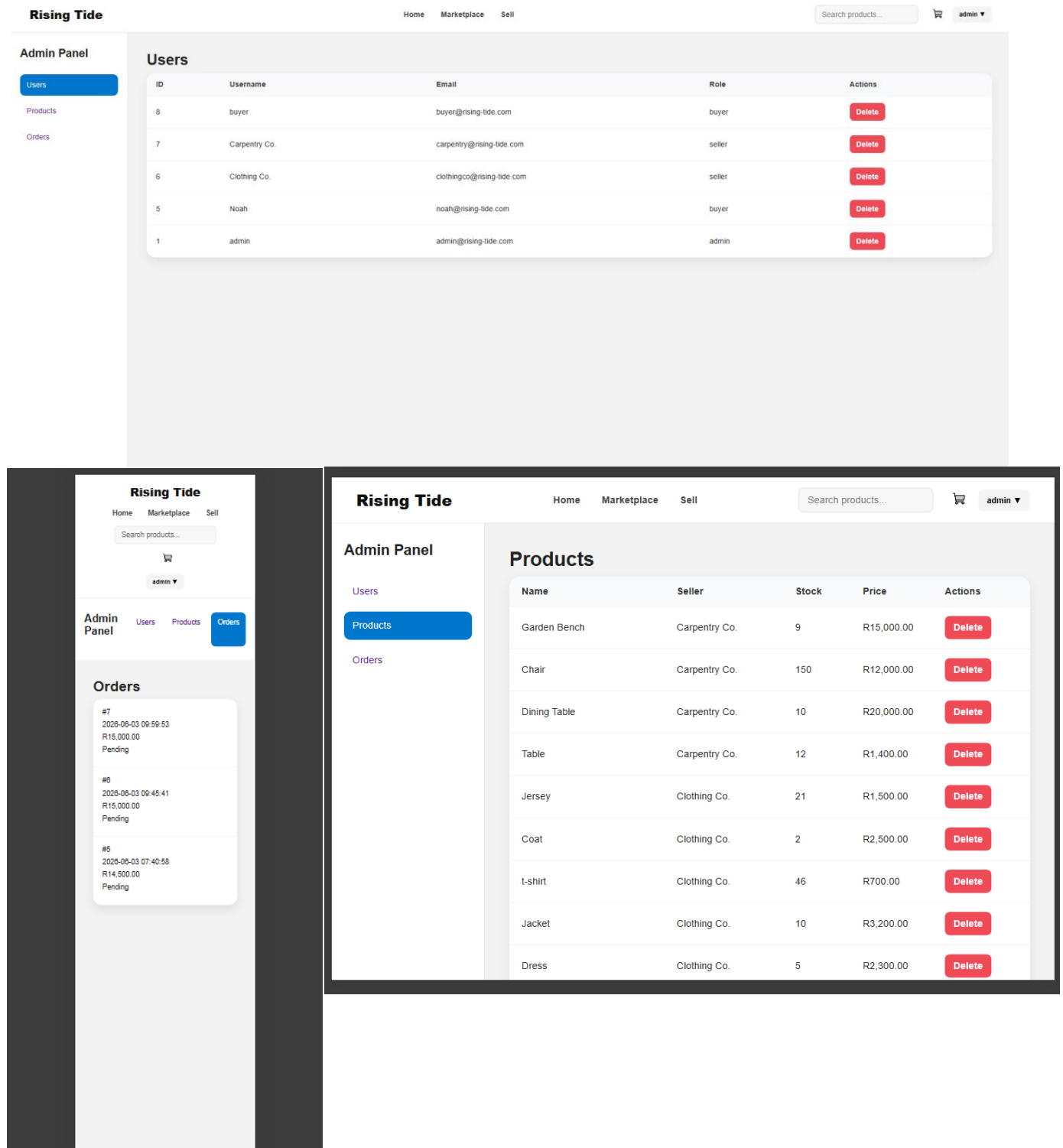


Figure 6: Admin dashboard with tabs for users, products and orders.

Design Diagrams

Class-Responsibility-Collaboration (CRC) Cards

User	
Responsibilities	Collaborators
• Register account log in • manage profile • list products for sale • add items to cart • place orders	• Product • Cart • Order

Product	
Responsibilities	Collaborators
• Store product information • display listings • manage stock levels • update product details	• User • Cart Item • Order Item

Cart	
Responsibilities	Collaborators
• Store selected products • update quantities • calculate cart contents	• User • Cart Item

Order	
Responsibilities	Collaborators
• Create orders • calculate totals • store order status • maintain order history	• User • OrderItem

Cart Item	
Responsibilities	Collaborators
• Link products to carts • store quantities • manage cart contents	• Cart • Product

Order Item	
Responsibilities	Collaborators
• Link products to orders • store quantities and prices at purchase time	• Order • Product

Enhanced Entity Relationship Diagram (EERD)

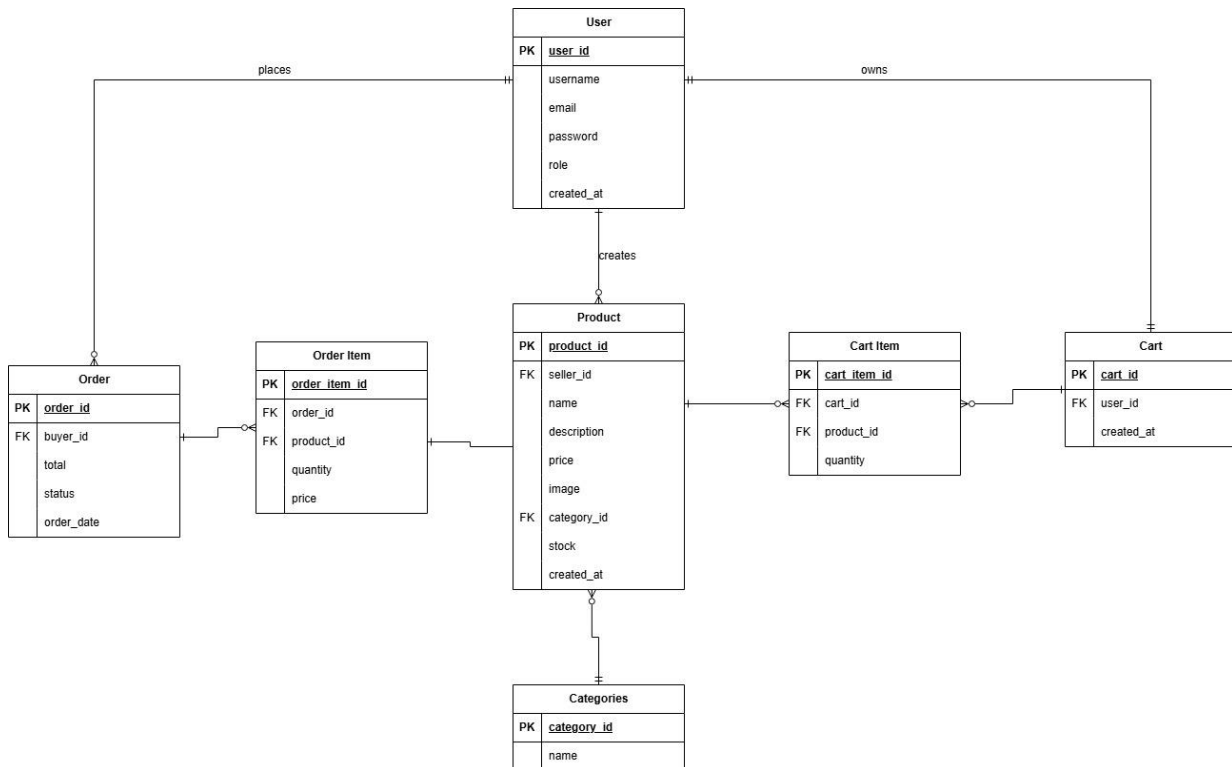
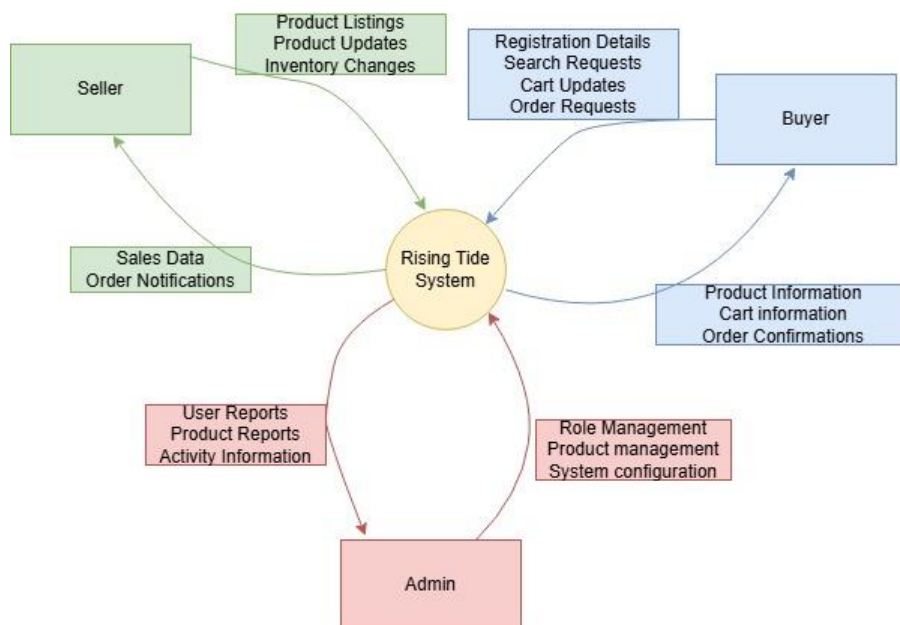


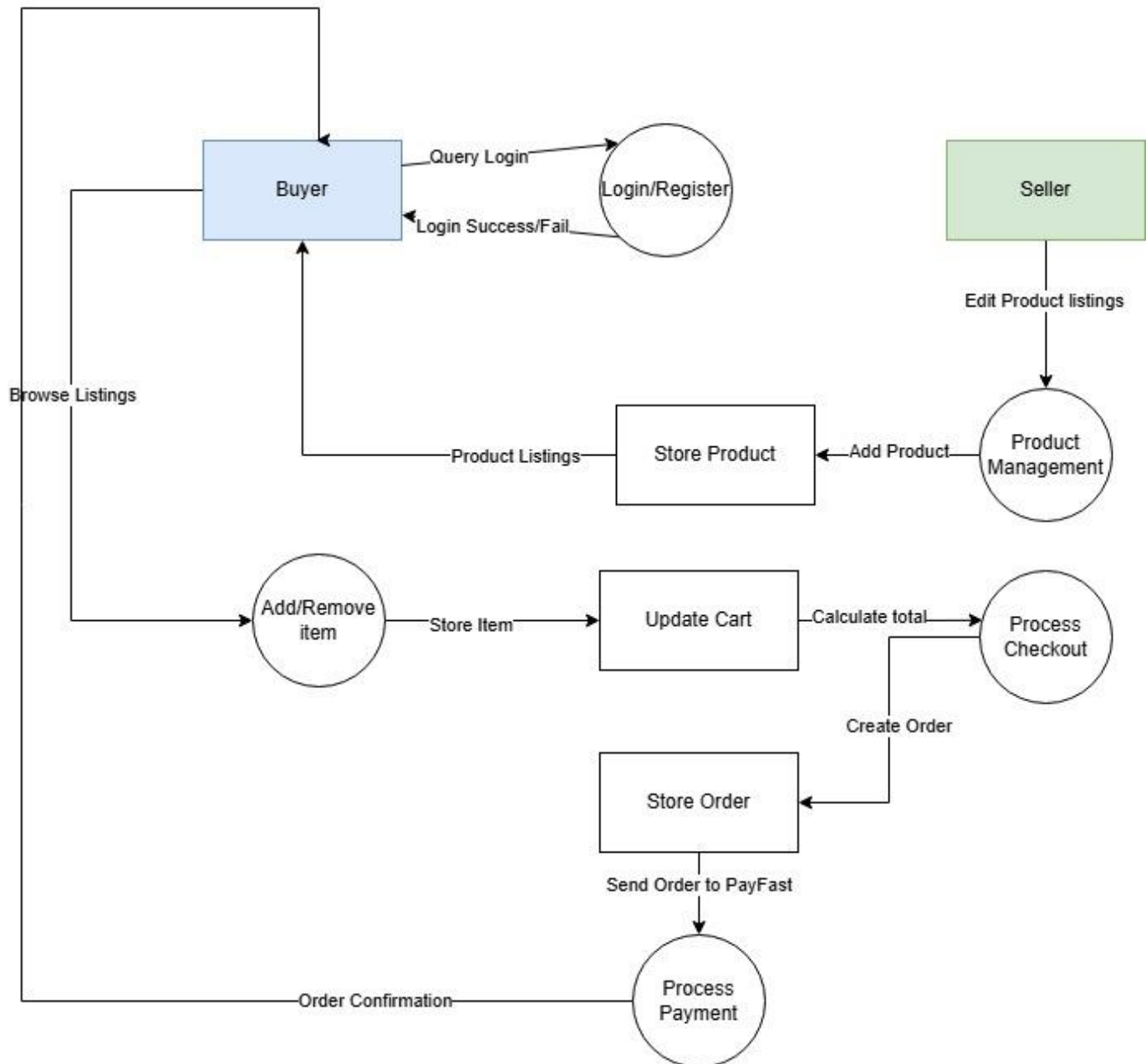
Figure 7: Illustrates the Rising Tide marketplace database with User as its central entity. Users create product listings and are linked to carts which contain cart items. Orders and order items help to manage purchases and store transactional data. Primary and foreign keys enforce the relationship between entities and ensure integrity and structure.

Context Diagram



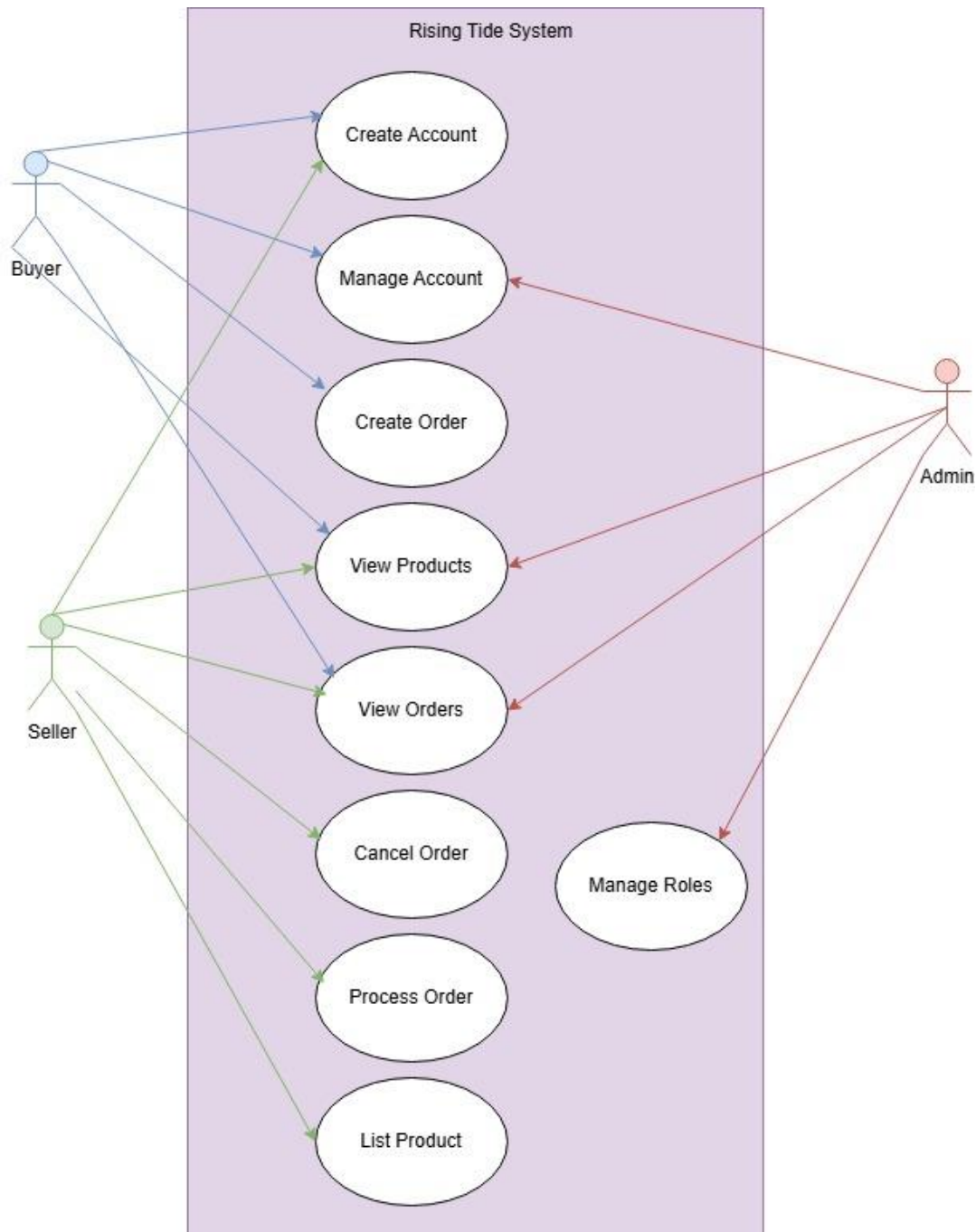
Data Flow Diagram (DFD)

This Data Flow Diagram displays how data moves through the Rising Tide system, showing how external entities like buyers and sellers engage with processes such as user authentication, product management, data storage and payment systems. It provides a clear overview of how information is processed, stored and transferred within the system to support the marketplace functionality.



Use Case Diagram

This diagram represents the functional requirements of the Rising Tide platform from a user's perspective. It shows different actors (e.g. Buyers, sellers and administrators) and the actions they can perform. These actions include creating accounts, listing products, managing user roles and making purchases. This assists in clarifying interactions between users and the system.



Database Design:

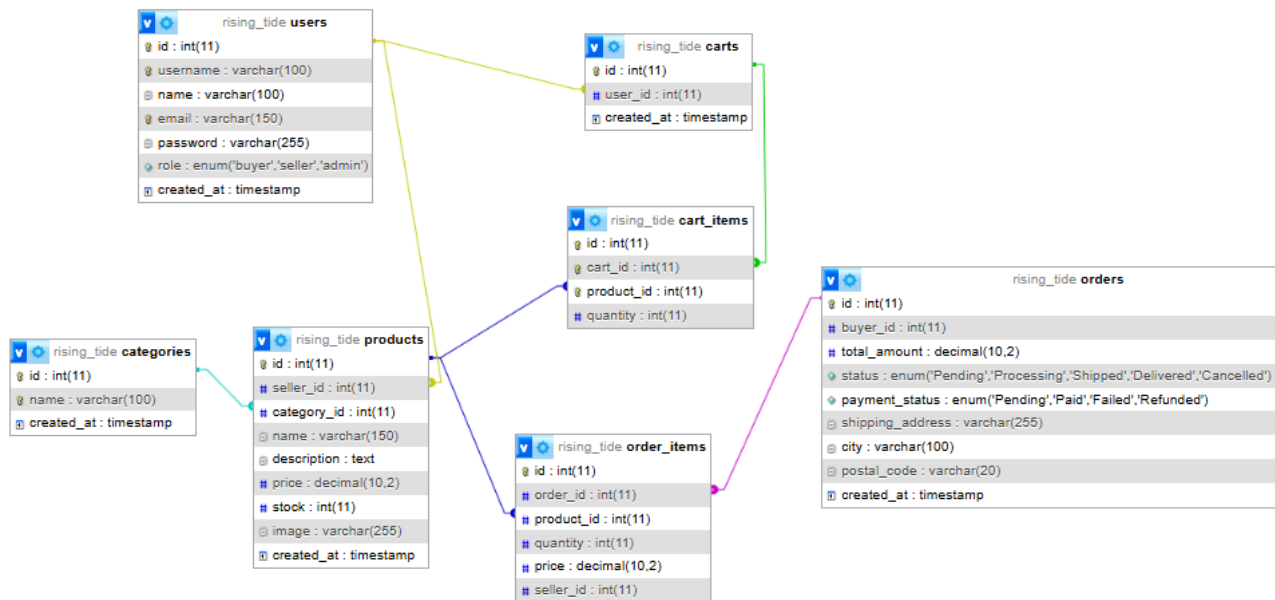


Figure 8: An Exported database schema from infinityfree's myphpadmin portal. It displays each entity of the database alongside Primary Key (PK) and Foreign Key (FK) relations between them.

Server: 127.0.0.1 Database: rising_tide

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers

Filters

Containing the word:

Table	Action	Rows	Type	Collation	Size	Overhead
☐ carts		4	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ cart_items		0	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ categories		20	InnoDB	utf8mb4_general_ci	32.0 KiB	-
☐ orders		1	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ order_items		2	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ products		10	InnoDB	utf8mb4_general_ci	48.0 KiB	-
☐ users		5	InnoDB	utf8mb4_general_ci	48.0 KiB	-
7 tables	Sum	42	InnoDB	utf8mb4_general_ci	304.0 KiB	0 B

☐ Check all With selected:

Print Data dictionary

Create new table

Table name Number of columns

Code Review

This section specifies the different web technologies utilised to create the rising tide platform and will include code snippets with short explanations to provide examples.

PHP

Snippet #1:

```
auth > process_login.php
1  <?php
2  session_start();
3  include('../config/db.php');
4
5  $email = $_POST['email'];
6  $password = $_POST['password'];
7
8  $stmt = $conn->prepare("SELECT * FROM users WHERE email = ?");
9  $stmt->bind_param("s", $email);
10 $stmt->execute();
11
12 $result = $stmt->get_result();
13 $user = $result->fetch_assoc();
14
15 if ($user && password_verify($password, $user['password'])) {
16
17     $_SESSION['user_id'] = $user['id'];
18     $_SESSION['username'] = $user['username'];
19     $_SESSION['role'] = $user['role'];
20
21     header("Location: ../user/dashboard.php");
22     exit();
23 } else {
24     echo "Invalid login credentials";
25 }
26
27 >>
```

Explanation: This PHP script processes the user authentication process for Rising Tide. It begins by starting a session (using `session_start()`) which allows user data to be stored across different pages after logging in. After that it includes the `db.php` file which creates a database connection to enable communication. The user's email and password are received from the HTML login form and matched with a corresponding user's information. The `bind_param()` function creates a prepared SQL statement helping to prevent against SQL injections. If the login credentials are valid, session variables such as `user_id`, `role` and `username` are created which help to enforce access control and identify users. If authentication fails, an error message displays 'Invalid login credentials'.

Snippet #2:

```
admin.php
1  <?php
2  include('includes/auth_check.php');
3  include('includes/header.php');
4  include('config/db.php');
5
6  if (!isset($_SESSION['role']) || $_SESSION['role'] !== 'admin') {
7      http_response_code(403);
8      die("Access denied");
9  }
10
11  $tab = $_GET['tab'] ?? 'users';
12
```

Explanation: This code forms the initial authentication check for the administrator dashboard. The first three include statements import the navigation header, css files, database connection and login authentication session variables. It then does a role-based access control check to see if the user in the active session has the admin role. If no role is found, an access denied message is displayed.

HTML

Snippet #1:

```
includes > header.php
7  <!DOCTYPE html>
8  <html lang="en">
9  <head>
10
11      <meta charset="UTF-8">
12
13      <meta
14          name="viewport"
15          content="width=device-width, initial-scale=1.0"
16      >
17
18      <title>Rising Tide</title>
19
20      <link rel="stylesheet" href="/rising-tide/assets/css/global.css">
21
22      <link rel="stylesheet" href="/rising-tide/assets/css/navbar.css">
23
24      <link rel="stylesheet" href="/rising-tide/assets/css/dashboard.css">
25
26      <link rel="stylesheet" href="/rising-tide/assets/css/forms.css">
27
28      <link rel="stylesheet" href="/rising-tide/assets/css/auth.css">
29
30      <link rel="stylesheet" href="/rising-tide/assets/css/marketplace.css">
31
32      <link rel="stylesheet" href="/rising-tide/assets/css/product.css">
33
34      <link rel="stylesheet" href="/rising-tide/assets/css/index.css">
35
36      <link rel="stylesheet" href="/rising-tide/assets/css/admin.css">
37
38  </head>
```

Explanation: This HTML snippet is found in the header.php file and displays the <head> section which includes links to every css file for the project. It also specifies the title of the website which displays in the tab of the browser as well as the specified language of English. This code is found specifically in the header because it will subsequently be imported into every other webpage on the Rising Tide platform, reducing the need to repeat code snippets on every page.

Snippet #2:

```
<form class="auth-form" action="process_login.php" method="POST">

  <div>
    <label>Email</label>

    <input
      type="email"
      name="email"
      placeholder="Enter your email"
      required
    >
  </div>

  <div>
    <label>Password</label>
    <input
      type="password"
      name="password"
      placeholder="*****"
      required
    >
  </div>
</form>
```

Explanation: This snippet displays part of the HTML login form. It displays multiple input fields, one for email and another for passwords. The password field type allows for the masking of user input so as to hide the input characters. The information input into these fields will be forwarded to the process_login.php file as variables to cross-check with the database for user authentication.

JavaScript

Snippet #1:

```
assets > js > JS script.js > showToast
1  function showToast(message, type = "success") {
2    const container = document.getElementById("toast-container");
3
4    const toast = document.createElement("div");
5    toast.classList.add("toast", type);
6    toast.innerText = message;
7
8    container.appendChild(toast);
9
10   setTimeout(() => {
11     toast.remove();
12   }, 2500);
13 }
```

Explanation: This JavaScript Function is commonly known as a 'Toast notification' and works to display temporary popup messages on the website when certain actions are done by the user. In the case of Rising Tide, toast notifications are used to display a pop-up message for when items are successfully added to cart. The function receives the container element using `getElementById`. After that it creates a new `<div>` element and assigns it the specified message. Once created the notification is added to the container using `appendChild()`. Finally, `setTimeout()` is used to remove the pop up after 2.5 seconds (or 2500 milliseconds).

CSS

Snippet #1:

```
.form-container button[type="submit"] {  
  padding: 1rem 2rem;  
  font-weight: 700;  
  background: ■ rgb(255, 255, 255);  
  color: ■ blueviolet;  
  cursor: pointer;  
  border-radius: 0.5rem;  
  border-bottom: 2px solid ■ blueviolet;  
  border-right: 2px solid ■ blueviolet;  
  border-top: 2px solid ■ white;  
  border-left: 2px solid ■ white;  
  transition-duration: 1s;  
  transition-property:  
    border-top, border-left, border-bottom, border-right, box-shadow;  
}  
  
.form-container button[type="submit"]:hover {  
  border-top: 2px solid ■ blueviolet;  
  border-left: 2px solid ■ blueviolet;  
  border-bottom: 2px solid ■ rgb(238, 103, 238);  
  border-right: 2px solid ■ rgb(238, 103, 238);  
  box-shadow:  
    ■ rgba(240, 46, 170, 0.4) 5px 5px,  
    ■ rgba(240, 46, 170, 0.3) 10px 10px,  
    ■ rgba(240, 46, 170, 0.2) 15px 15px;  
}
```

Explanation: This CSS styles snippet features the unique styling attached to the submit buttons of every form on Rising Tide. The font-weight is bold for better visibility creating a strong call to action for every button. Each button has a border radius set to 0.5 rem to give a softer and more modern appearance. Importantly, when hovering over the buttons, a multi-layered, vibrant shadow effect is created. This gives the user immediate visual feedback to help improve engagement with the site.

MySQL

Snippet #1:

```
$usersResult = $conn->query("
    SELECT *
    FROM users
    ORDER BY id DESC
");

$productsResult = $conn->query("
    SELECT p.*, u.username AS seller_name
    FROM products p
    LEFT JOIN users u ON p.seller_id = u.id
    ORDER BY p.created_at DESC
");

$ordersResult = $conn->query("
    SELECT *
    FROM orders
    ORDER BY created_at DESC
");
```

Explanation: This snippet includes three SQL queries within the administrator's dashboard page. All three queries are used to retrieve important information from the database that involves user activity and information. The first query retrieves every record from the users table (except passwords) and orders it based on the latest user created. The second query retrieves product information from the products table and uses a LEFT JOIN with the user's table. This matches the seller ids with their respective products providing a comprehensive breakdown of every listed product in the database alongside their respective seller's information. The third query retrieves all completed orders and arranges them from most recent. This allows administrators to view order records in case of a dispute.

Snippet #2:

```
1 CREATE TABLE `orders` (  
2   `id` int(11) NOT NULL AUTO_INCREMENT,  
3   `buyer_id` int(11) NOT NULL,  
4   `total_amount` decimal(10,2) NOT NULL DEFAULT 0.00,  
5   `status` enum('Pending','Processing','Shipped','Delivered','Cancelled') DEFAULT 'Pending',  
6   `payment_status` enum('Pending','Paid','Failed','Refunded') DEFAULT 'Pending',  
7   `shipping_address` varchar(255) DEFAULT NULL,  
8   `city` varchar(100) DEFAULT NULL,  
9   `postal_code` varchar(20) DEFAULT NULL,  
10  `created_at` timestamp NOT NULL DEFAULT current_timestamp(),  
11  PRIMARY KEY (`id`),  
12  KEY `buyer_id` (`buyer_id`)|
```

Explanation: This is an example of the MySQL statement that created the orders table. This table is responsible for storing the buyer's ID, order amount, shipping details and current order status. The status and payment_status fields utilise ENUM which restricts their values to predefined states such as pending or delivered. This preserves the integrity of the database and simplifies order management.

Conclusion

The Rising Tide platform was created in response to the growing need for a secure and trusted C2C marketplace focusing specifically on the informal sector of South Africa. Throughout its development cycle, this project successfully delivered a minimum viable product (MVP) that showcases the core system features needed such as product listing, role-based access controls (RBAC) and structured database storage. Overall, Rising Tide demonstrates that a digital marketplace can empower users from small township traders to established vendors by providing a more reliable online commerce solution. By doing so, it aligns with the guiding principle of 'a rising tide lifts all boats' by promoting inclusive participation and contributing to a broader digital transformation in South Africa. This solution can be further expanded upon in the future with additional features such as sales analytics, payment integration and verified review and report functionality.